THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the nephron, glucose is reabsorbed in the: 1

(a) bladder **(b)** glomerulus **(c)** ureter **(d)** tubule

2. Bile is alkaline to: 1

(a) kill bacteria
(b) digest starch
(c) make the medium suitable for pancreatic enzymes
(d) digest protein

3. The rings of cartilage in the trachea prevent it from: 1

(a) bleeding **(b)** collapsing **(c)** drying **(d)** contracting

4. The urine contains mainly: 1

(a) urea, water and salts **(b)** glucose, water
(c) proteins **(d)** blood cells

5. Fats are broken down into small globules by: 1

(a) pepsin **(b)** bile salts **(c)** trypsin **(d)** amylase

6. The oxygen-rich blood in a human heart is found in: 1

(a) left atrium and left ventricle **(b)** left atrium and right ventricle
(c) right atrium and right ventricle **(d)** right atrium and left ventricle

7. Which is not an excretory product in plants? 1

(a) gum **(b)** resin **(c)** latex **(d)** chlorophyll

8. The finger-like structures in the small intestine are: 1

(a) stomata **(b)** villi **(c)** nephrons **(d)** alveoli

9. The thick wall of the left ventricle is due to: 1

(a) pumping blood to the whole body **(b)** pumping to lungs
(c) valve action **(d)** receiving blood

10. Platelets help in: 1

(a) digestion **(b)** carrying oxygen **(c)** blood clotting **(d)** fighting germs

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State the role of root pressure and transpiration pull in transport of water. 2

12. Differentiate between ingestion and egestion. 2

13. Why does blood pressure record two readings? 2

14. Why is the movement of materials through phloem called active transport? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how the lungs ensure sufficient oxygen supply during exercise. 3

16. Compare transpiration and translocation. 3

17. Describe how guard cells control the stomatal opening in light and darkness. 3

OR

Differentiate between aerobic respiration in humans and fermentation in yeast. 3

18. Explain how the nephron is adapted to filter blood and reabsorb useful substances. 3

OR

What are the products of digestion of carbohydrates, proteins and fats? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe transport in plants through the xylem and the factors affecting transpiration. 5

OR

Explain nutrition in Amoeba and Paramecium. 5

20. Explain the formation of urine in a nephron and why urine production varies. 5

OR

Explain the importance of breathing rate, haemoglobin and heart in providing energy to cells. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND